

## Problem Set 2

QiLiu 25340516

October 22, 2025

### Question 1: Political Science — Chi-Square Test

Observed counts

| Class \ Outcome | Not Stopped | Bribe | Warning | Row total |
|-----------------|-------------|-------|---------|-----------|
| Upper           | 14          | 6     | 7       | 27        |
| Lower           | 7           | 7     | 1       | 15        |
| Column total    | 21          | 13    | 8       | 42        |

(a) **Compute the chi-square statistic (show the arithmetic).**

Let  $n = 42$ . Expected counts under independence are

$$E_{ij} = \frac{(\text{row}_i)(\text{col}_j)}{n}.$$

Hence

$$\begin{aligned} E_{\text{Upper,NS}} &= \frac{27 \times 21}{42} = 13.5, & E_{\text{Upper,Bribe}} &= \frac{27 \times 13}{42} = 8.3571429, & E_{\text{Upper,Warn}} &= \frac{27 \times 8}{42} = 5.1428571, \\ E_{\text{Lower,NS}} &= \frac{15 \times 21}{42} = 7.5, & E_{\text{Lower,Bribe}} &= \frac{15 \times 13}{42} = 4.6428571, & E_{\text{Lower,Warn}} &= \frac{15 \times 8}{42} = 2.8571429. \end{aligned}$$

The Pearson chi-square statistic is

$$\begin{aligned} \chi^2 &= \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \\ &= \frac{(14 - 13.5)^2}{13.5} + \frac{(6 - 8.3571429)^2}{8.3571429} + \frac{(7 - 5.1428571)^2}{5.1428571} \\ &\quad + \frac{(7 - 7.5)^2}{7.5} + \frac{(7 - 4.6428571)^2}{4.6428571} + \frac{(1 - 2.8571429)^2}{2.8571429} \\ &= 0.0137 + 0.6648 + 0.6519 + 0.0333 + 1.2907 + 1.1368 \\ &= \mathbf{3.7912} \text{ (rounded)}. \end{aligned}$$

*R verification:*

```

1 obs <- matrix(c(14,6,7, 7,7,1), nrow=2, byrow=TRUE)
2 rownames(obs) <- c("Upper","Lower")
3 colnames(obs) <- c("NotStopped","Bribe","Warning")
4 rowSums <- rowSums(obs); colSums <- colSums(obs); total <- sum(obs)
5 expected <- outer(rowSums,colSums)/total
6 chi2 <- sum((obs-expected)^2/expected); chi2

```

Listing 1: Q1(a) R verification

- (b) **Compute the p-value at  $\alpha = 0.10$  and conclude.**

Degrees of freedom:  $(r-1)(c-1) = (2-1)(3-1) = 2$ . Right-tail p-value:

$$p = \Pr\{\chi_{(2)}^2 \geq 3.7912\} = \mathbf{0.1502}.$$

Since  $p = 0.1502 > 0.10$ , we **fail to reject** the null hypothesis of independence.

R: `chisq.test(obs, correct=FALSE)` returns  $X^2 = 3.7912$ ,  $df = 2$ ,  $p = 0.1502$  (warning appears because some  $E_{ij} < 5$ ).

- (c) **Compute standardized residuals and fill the table.**

Pearson residuals  $r_{ij} = (O_{ij} - E_{ij})/\sqrt{E_{ij}}$ :

|       | Not Stopped | Bribe   | Warning |
|-------|-------------|---------|---------|
| Upper | 0.1361      | -0.8154 | 0.8189  |
| Lower | -0.1826     | 1.0939  | -1.0987 |

Adjusted standardized residuals:

|       | Not Stopped | Bribe   | Warning |
|-------|-------------|---------|---------|
| Upper | 0.3220      | -1.6420 | 1.5230  |
| Lower | -0.3220     | 1.6420  | -1.5230 |

R snippet:

```

1 pearson_res <- (obs - expected) / sqrt(expected)
2 row_prop <- rowSums/total; col_prop <- colSums/total
3 adj_res <- matrix(NA, nrow=2, ncol=3)
4 for(i in 1:2){
5   for(j in 1:3){
6     adj_res[i,j] <- (obs[i,j]-expected[i,j]) /
7       sqrt(expected[i,j]*(1-row_prop[i])*(1-col_prop[j]))
8   }
9 }

```

Listing 2: Q1(c) residuals in R

(d) **Interpretation using the residuals.**

As a rule of thumb,  $|r| > 1.96$  suggests a cell-level deviation at the 5% level. Here, all adjusted residuals are  $|r| < 2$  (largest  $\approx 1.64$ ), so no cell departs significantly from independence. This aligns with the overall chi-square test ( $p = 0.1502$ ): there is no statistically significant association between class and outcome.

## Question 2: Economics — Regression Analysis

We analyze the effect of political reservation for women on drinking water facilities.

(a) **State the hypotheses (two-tailed).**

$$H_0 : \beta = 0 \quad (\text{reservation has no effect})$$

$$H_1 : \beta \neq 0 \quad (\text{reservation affects the outcome})$$

Model:  $\text{water}_i = \alpha + \beta \text{reserved}_i + \varepsilon_i$ , where  $\text{reserved} \in \{0, 1\}$ .

(b) **Estimate the model in R (OLS and robust SE).**

```
1 url <- "https://raw.githubusercontent.com/kosukeimai/qss/master/
  PREDICTION/women.csv"
2 women <- read.csv(url)
3
4 model <- lm(water ~ reserved, data = women)
5 summary(model)
6
7 if (!require("sandwich")) install.packages("sandwich")
8 if (!require("lmtest"))   install.packages("lmtest")
9 library(sandwich); library(lmtest)
10 coeftest(model, vcov = vcovHC(model, type = "HC1"))
```

Listing 3: Q2(b) R code

**Numerical results from your R session:**

- OLS:  $\hat{\beta} = 9.2524$ ,  $\text{SE} = 3.948$ ,  $t = 2.344$ ,  $p = 0.0197$ ; Residual  $\text{SE} = 33.45$  (320 df);  $R^2 = 0.01688$ ; adj.  $R^2 = 0.0138$ ;  $F(1, 320) = 5.493$ ,  $p = 0.0197$ .
- Robust (HC1):  $\hat{\beta} = 9.2524$ ,  $\text{SE} = 5.0932$ ,  $t = 1.8166$ ,  $p = 0.0702$ .

(c) **Interpret the coefficient and statistical significance.**

$\hat{\beta} = 9.25$  means villages with reserved seats for women have on average  $\approx 9.25$  more new/repaired water facilities than non-reserved villages, holding the specification fixed. Under conventional OLS SEs this is statistically significant at 5% ( $p = 0.0197$ ). Using heteroskedasticity-robust SEs (HC1), the effect remains positive but is only marginally significant at 10% ( $p = 0.0702$ ). Model fit is weak ( $R^2 \approx 0.017$ ), so practical significance should be interpreted cautiously.